

**MATH 3551 A — Abstract Algebra I**  
**Practice Final Examination — SOLUTIONS**

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**Problem 1.** (60 points) **True or false; prove or disprove.**

- (a) **TRUE.**  $(1\ 2\ 3)(4\ 5\ 6\ 7)$  has order  $\text{lcm}(3, 4) = 12$  and uses  $3 + 4 = 7$  points.
- (b) **FALSE.** A homomorphism  $\mathbb{Z}/12\mathbb{Z} \rightarrow \mathbb{Z}/8\mathbb{Z}$  is determined by the image  $x$  of  $\bar{1}$ , subject to  $12x \equiv 0 \pmod{8}$ , i.e.  $4x \equiv 0 \pmod{8}$ , i.e.  $x$  even. Taking  $x = \bar{2}$  gives a nontrivial homomorphism (with image of order 4).
- (c) **TRUE.**  $|(\mathbb{Z}/18\mathbb{Z})^\times| = \varphi(18) = 6$ , and  $\bar{5}$  has order 6:  $5, 25 \equiv 7, 35 \equiv 17, 85 \equiv 13, 65 \equiv 11, 55 \equiv 1$ .
- (d) **TRUE.** This is D&F §4.2 Corollary 5: act on the  $p$  cosets to get  $\pi: G \rightarrow S_p$  with  $K = \ker \pi \leq H$ ; then  $|G : K|$  divides both  $p!$  and  $|G|$ , and since  $p$  is the smallest prime divisor of  $|G|$  this forces  $|G : K| = p$ , i.e.  $K = H$ .
- (e) **FALSE.**  $105 = 3 \cdot 5 \cdot 7$ , and there is a nonabelian group  $\mathbb{Z}/5\mathbb{Z} \times (\mathbb{Z}/7\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z})$  of that order (the semidirect product exists because  $3 \mid |\text{Aut}(\mathbb{Z}/7\mathbb{Z})| = 6$ ).
- (f) **TRUE.** Choose the six moved points in  $\binom{8}{6} = 28$  ways and arrange them in a cycle in  $5! = 120$  ways:  $28 \cdot 120 = 3360$ .
- (g) **TRUE.** An automorphism of  $V_4$  permutes its three elements of order 2, and every permutation of them extends to an automorphism (any bijection between two-element generating sets of  $V_4$  is an automorphism); so  $\text{Aut}(V_4) \cong S_3$ .
- (h) **TRUE.**  $|A_5| = 60 = 2^2 \cdot 3 \cdot 5$  and  $n_5 = 6$ , so the normalizer of a Sylow 5-subgroup has order  $60/6 = 10$ . (It is the dihedral group  $D_{10}$  generated by a 5-cycle and a double transposition inverting it.)
- (i) **FALSE.**  $A_4$  has order 12 and  $Z(A_4) = 1$ , so its upper central series never leaves 1:  $A_4$  is not nilpotent.
- (j) **FALSE.**  $36 = 2^2 \cdot 3^2$  gives  $n_3 \in \{1, 4\}$ . If the group were simple then  $n_3 = 4$ , and the conjugation action on the four Sylow 3-subgroups gives  $G \rightarrow S_4$  with kernel a proper normal subgroup, hence trivial by simplicity; but then  $36 \nmid 24$ , which is false.
- (k) **TRUE.**  $|Q_8| = 8 = 2^3$ , and every finite  $p$ -group is nilpotent.
- (l) **FALSE.**  $Z(A_4) = 1$  (see item (i)).
- (m) **TRUE.** They are  $\mathbb{Z}/8\mathbb{Z}$ ,  $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ ,  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ ,  $D_8$  and  $Q_8$ .

- (n) **FALSE.**  $D_8$  is a nonabelian group of order  $2^3$ ; for odd  $p$  the Heisenberg group of upper unitriangular  $3 \times 3$  matrices over  $\mathbb{F}_p$  is nonabelian of order  $p^3$ .
- (o) **TRUE.**  $70 = 2 \cdot 5 \cdot 7$ ; the four groups are  $\mathbb{Z}/70\mathbb{Z}$ ,  $\mathbb{Z}/7\mathbb{Z} \times D_{10}$ ,  $\mathbb{Z}/5\mathbb{Z} \times D_{14}$  and  $D_{70}$ .
- (p) **FALSE.**  $D_8$  is nonabelian, but  $Z(D_8) = \langle r^2 \rangle$  and  $D_8/\langle r^2 \rangle$  has order 4, hence is abelian. (What is true is that  $G/Z(G)$  is never nontrivial *cyclic*.)
- (q) **TRUE.** With D&F's convention  $[a, b] = a^{-1}b^{-1}ab$  we get  $[s, r] = s^{-1}r^{-1}sr = (sr^{-1}s)r = r \cdot r = r^2$ , so  $\langle r^2 \rangle \leq D'_8$ ; and  $D_8/\langle r^2 \rangle$  has order 4 and is abelian, so  $D'_8 \leq \langle r^2 \rangle$ .
- (r) **TRUE.**  $A'_n \trianglelefteq A_n$  and  $A_n$  is simple for  $n \geq 5$ , so  $A'_n$  is 1 or  $A_n$ ; it is not 1, since  $A_n$  is nonabelian.
- (s) **TRUE.** If  $\overline{G} = G/N$  then  $\gamma^i(\overline{G})$  is the image of  $\gamma^i(G)$ , so the lower central series of  $\overline{G}$  reaches 1 at least as fast as that of  $G$ .
- (t) **FALSE.** The number of abelian groups of order  $2^5$  is the number of partitions of 5, which is 7, not 6.

**Grading note.** Three points each: one for the circle, two for the reason. The discriminating items are (b), (d), (h), (p) and (t); (i) and (l) test the same fact twice on purpose, so a student who answers them inconsistently should be looked at. Accept any correct counterexample.

**Problem 2.** (20 points) **A permutation.**

- (a)  $|\tau| = \text{lcm}(4, 2, 3) = 12$ .
- (b)  $100 \equiv 0 \pmod{4}$ ,  $100 \equiv 0 \pmod{2}$ ,  $100 \equiv 1 \pmod{3}$ , so only the 3-cycle survives:  $\tau^{100} = (689)$ .
- (c) A  $k$ -cycle is a product of  $k - 1$  transpositions, so  $\tau$  is a product of  $3 + 1 + 2 = 6$  transpositions:  $\tau$  is **even**.
- (d) Yes: both have cycle type  $(4, 3, 2)$ , and permutations of  $S_9$  are conjugate exactly when they have the same cycle type. To build  $\pi$ , write the two permutations with their cycles in matching order and read off the correspondence:

$$\tau = (1247)(689)(35), \quad \rho = (1234)(567)(89),$$

so take  $\pi(1) = 1$ ,  $\pi(2) = 2$ ,  $\pi(4) = 3$ ,  $\pi(7) = 4$ ,  $\pi(6) = 5$ ,  $\pi(8) = 6$ ,  $\pi(9) = 7$ ,  $\pi(3) = 8$ ,  $\pi(5) = 9$ ; in cycle notation

$$\pi = (3865974).$$

Then  $\pi\tau\pi^{-1} = (\pi1\pi2\pi4\pi7)(\pi6\pi8\pi9)(\pi3\pi5) = (1234)(567)(89) = \rho$ . ✓

- (e) The class of  $\tau$  has  $\frac{9!}{4 \cdot 3 \cdot 2} = \frac{362880}{24} = 15120$  elements, so  $|C_{S_9}(\tau)| = 362880/15120 = 24$ .

**Grading note.** Part (d) is where the credit is: the conjugating permutation must be produced, and it must be checked or at least visibly read off from aligned cycle decompositions. Award 2 points for “same cycle type, hence conjugate” and 3 for a correct  $\pi$ . Any correct  $\pi$  is fine — there are 24 of them, one for each element of the centralizer times this one.

**Problem 3.** (20 points) **A group of order 204.**

- (a)  $n_{17} \mid 12$  and  $n_{17} \equiv 1 \pmod{17}$ . The divisors of 12 are 1, 2, 3, 4, 6, 12, all less than 17, so only  $n_{17} = 1$  is congruent to 1. Hence the Sylow 17-subgroup  $P$  is unique, therefore normal.
- (b) Let  $Q \in \text{Syl}_3(G)$ , of order 3. Since  $P \trianglelefteq G$ ,  $PQ$  is a subgroup;  $|P \cap Q|$  divides  $\gcd(17, 3) = 1$ , so  $|PQ| = 17 \cdot 3 = 51$ .
- (c) First, every group  $K$  of order  $12 = 2^2 \cdot 3$  has a normal Sylow subgroup:  $n_3 \in \{1, 4\}$ , and if  $n_3 = 4$  then the four Sylow 3-subgroups contribute  $4 \cdot 2 = 8$  elements of order 3, leaving  $12 - 8 = 4$  elements, which must therefore be the unique Sylow 2-subgroup, so  $n_2 = 1$ .

Now  $P \trianglelefteq G$  with  $P \cong \mathbb{Z}/17\mathbb{Z}$  abelian, and  $G/P$  has order  $204/17 = 12$ . By the previous paragraph  $G/P$  has a normal subgroup  $M/P$  with  $M/P$  of order 3 or 4, and the corresponding quotients are of order 4 or 3; in either case we obtain a chain

$$1 \trianglelefteq P \trianglelefteq M \trianglelefteq G$$

with quotients  $P \cong \mathbb{Z}/17\mathbb{Z}$ ,  $M/P$  of order 3 or 4, and  $G/M$  of order 4 or 3. Groups of order 3 and of order 4 are abelian, so all the factors are abelian and  $G$  is solvable. (Equivalently:  $P$  is solvable and  $G/P$  of order 12 is solvable, and an extension of a solvable group by a solvable group is solvable.)

- (d)  $P$  is the unique subgroup of order 17, and every element of order 17 generates a subgroup of order 17, hence lies in  $P$ . So the elements of order 17 are the  $17 - 1 = \boxed{16}$  non-identity elements of  $P$ .

**Grading note.** Part (c) is worth 7 and can be done either by exhibiting the chain or by citing the extension property of solvability (D&F §3.4 #8) — but in either case the order-12 lemma has to be established, and that is 4 of the 7 points. Expected error in (a): forgetting to check that all divisors of 12 are below 17, and instead writing “ $n_{17} = 1$  or 18”; 18 is not a divisor of 12.

**Problem 4.** (20 points) **The groups of order 55.**

- (a)  $55 = 5 \cdot 11$ , and  $n_{11} \mid 5$  with  $n_{11} \equiv 1 \pmod{11}$  forces  $n_{11} = 1$ . So the Sylow 11-subgroup  $P \cong \mathbb{Z}/11\mathbb{Z}$  is normal.
- (b) Let  $Q \in \text{Syl}_5(G)$ , so  $Q \cong \mathbb{Z}/5\mathbb{Z}$ . Then  $P \cap Q = 1$  (its order divides  $\gcd(11, 5) = 1$ ) and  $|PQ| = 55 = |G|$ , so  $PQ = G$ . Together with  $P \trianglelefteq G$  this says  $G$  is the internal semidirect product  $P \rtimes_{\varphi} Q$ , where  $\varphi(q)$  is conjugation by  $q$ .
- (c)  $\text{Aut}(P) \cong (\mathbb{Z}/11\mathbb{Z})^{\times}$  is cyclic of order 10. A homomorphism  $\varphi: \mathbb{Z}/5\mathbb{Z} \rightarrow \mathbb{Z}/10\mathbb{Z}$  sends a generator to an element of order dividing 5, and a cyclic group of order 10 has exactly one subgroup of order 5, hence exactly five such elements. So there are five homomorphisms  $\varphi$ .
- $\varphi$  trivial gives  $G \cong \mathbb{Z}/11\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}/55\mathbb{Z}$ .
  - The four nontrivial  $\varphi$  all give isomorphic groups: if  $\varphi_1$  and  $\varphi_2$  are nontrivial then they have the same image (the unique subgroup of order 5 of  $\text{Aut}(P)$ ), so  $\varphi_2 = \varphi_1 \circ \alpha$  for some  $\alpha \in \text{Aut}(Q)$ , and precomposing with an automorphism of  $Q$  yields isomorphic semidirect products.

So up to isomorphism there are exactly **two** groups of order 55: the cyclic one and one nonabelian one.

- (d) Let  $G$  be the nonabelian one.  $Z(G) \trianglelefteq G$ , so  $|Z(G)| \in \{1, 5, 11, 55\}$ . If  $|Z(G)| = 11$  then  $P \leq Z(G)$  and the conjugation action of  $Q$  on  $P$  is trivial, making  $G$  abelian; if  $|Z(G)| = 5$  then  $G/Z(G)$  has order 11, hence is cyclic, hence  $G$  is abelian; and  $|Z(G)| = 55$  says  $G$  is abelian outright. So  $Z(G) = 1$ . For  $n_5$ :  $n_5 \mid 11$  and  $n_5 \equiv 1 \pmod{5}$ , so  $n_5 \in \{1, 11\}$ ; if  $n_5 = 1$  then  $Q \trianglelefteq G$  too and  $G \cong P \times Q$  would be abelian. Hence  $n_5 = 11$ .

**Grading note.** As on the real exam, the trap in (c) is “five homomorphisms, so five groups”. Award 2 points for  $\text{Aut}(\mathbb{Z}/11\mathbb{Z}) \cong \mathbb{Z}/10\mathbb{Z}$ , 3 for the count of five homomorphisms, 3 for the argument that the four nontrivial ones give one isomorphism class. In (d), the elimination of the three nontrivial possibilities for  $|Z(G)|$  is worth 2 and  $n_5 = 11$  worth 2.

**Problem 5.** (20 points) **Abelian groups.**

- (a)  $36 = 2^2 \cdot 3^2$ ,  $50 = 2 \cdot 5^2$ ,  $8 = 2^3$ , so  $|A| = 36 \cdot 50 \cdot 8 = 14400 = 2^6 \cdot 3^2 \cdot 5^2$ . The elementary divisors are

$$\underbrace{8, 4, 2}_{p=2}, \quad \underbrace{9}_{p=3}, \quad \underbrace{25}_{p=5}.$$

Three invariant factors; aligning from the largest and multiplying down the columns gives  $8 \cdot 9 \cdot 25 = 1800$ , then 4, then 2:

$$2 \mid 4 \mid 1800, \quad A \cong \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_{1800}$$

(check:  $2 \cdot 4 \cdot 1800 = 14400$ ).

- (b)  $p(6)p(2)p(2) = 11 \cdot 2 \cdot 2 = \boxed{44}$ .

- (c) The largest order of an element is the exponent, i.e. the largest invariant factor, 1800. Since  $1800 = 2^3 \cdot 3^2 \cdot 5^2$ :  $200 = 2^3 \cdot 5^2$  divides 1800, so **yes**, there is an element of order 200;  $16 = 2^4$  does not divide 1800, so **no** element of order 16.
- (d) ( $\Rightarrow$ ) If  $A$  is cyclic of order  $n = \prod p_i^{e_i}$ , then its Sylow  $p_i$ -subgroup is the unique subgroup of order  $p_i^{e_i}$  of a cyclic group, hence cyclic.
- ( $\Leftarrow$ ) Suppose each Sylow subgroup  $A_i$  is cyclic, say of order  $p_i^{e_i}$ , and pick a generator  $a_i$ . By the decomposition of a finite abelian group into its Sylow subgroups,  $A \cong A_1 \times \cdots \times A_k$ . The element  $a = a_1 a_2 \cdots a_k$  has order  $\text{lcm}(p_1^{e_1}, \dots, p_k^{e_k}) = \prod p_i^{e_i} = |A|$ , since the  $p_i^{e_i}$  are pairwise coprime. Hence  $A = \langle a \rangle$  is cyclic.

**Grading note.** The alignment rule in (a) is the whole skill: 8, 4, 2 against 9 and 25, padded with 1s. A common error is to produce two invariant factors rather than three. In (c) the phrase “the exponent is the largest invariant factor” should appear; a student who answers 14400 has confused order with exponent. Part (d) may quote the Sylow decomposition of a finite abelian group, but the coprime-lcm step must be there.

**Problem 6.** (20 points) **Solvable and nilpotent.**

- (a)  $\gamma^1(D_8) = D_8$ . Next  $\gamma^2(D_8) = [D_8, D_8] = \langle r^2 \rangle$  (because  $[s, r] = s^{-1}r^{-1}sr = r^2$ , and  $D_8/\langle r^2 \rangle$  has order 4, hence is abelian). Finally  $\gamma^3(D_8) = [D_8, \langle r^2 \rangle] = 1$ , since  $r^2 \in Z(D_8)$ . So the lower central series is

$$D_8 \geq \langle r^2 \rangle \geq 1,$$

and  $D_8$  is nilpotent of class 2.

- (b)  $A_4^{(1)} = A'_4 = V_4$ : indeed  $A_4/V_4$  has order 3, hence is abelian, so  $A'_4 \leq V_4$ ; and  $[(1\ 2\ 3), (1\ 2)(3\ 4)] = (1\ 4)(2\ 3) \neq 1$  lies in  $A'_4$ , so  $A'_4 \neq 1$ , and  $A'_4 \trianglelefteq A_4$  with  $V_4$  the only normal subgroup of order 4 forces  $A'_4 = V_4$  (the only proper nontrivial normal subgroup of  $A_4$  is  $V_4$ ). Then  $A_4^{(2)} = V'_4 = 1$  since  $V_4$  is abelian. The derived series is  $A_4 \triangleright V_4 \triangleright 1$ , so  $A_4$  is solvable of derived length 2.
- (c) In  $D_{2n} = \langle r, s \rangle$  the subgroup  $\langle r \rangle$  is cyclic of order  $n$  and has index 2, hence is normal, and  $D_{2n}/\langle r \rangle \cong \mathbb{Z}/2\mathbb{Z}$ . So  $1 \trianglelefteq \langle r \rangle \trianglelefteq D_{2n}$  is a chain with abelian quotients ( $\langle r \rangle \cong \mathbb{Z}/n\mathbb{Z}$  and  $\mathbb{Z}/2\mathbb{Z}$ ), and  $D_{2n}$  is solvable.
- (d)  $Z(A_4) = 1$ : a nonidentity element of  $A_4$  is either a 3-cycle or a double transposition, and neither is central — e.g.  $(1\ 2\ 3)$  does not commute with  $(1\ 2)(3\ 4)$ , and  $(1\ 2)(3\ 4)$  does not commute with  $(1\ 2\ 3)$ . So the upper central series of  $A_4$  is  $1 = Z_0 = Z_1 = \cdots$  and never reaches  $A_4$ :  $A_4$  is not nilpotent. (Equivalently: a finite nilpotent group has all Sylow subgroups normal, but  $n_3(A_4) = 4$ .)

**Grading note.** Part (b): 3 points for  $A'_4 \leq V_4$  (via the abelian quotient), 2 for the reverse. Part (d): either the centre argument or the Sylow argument is worth full credit, but “ $A_4$  is not abelian so it is not nilpotent” is false reasoning and earns 0 —  $D_8$  is nonabelian and nilpotent. In (c) watch for students who try to prove  $\langle r \rangle$  normal by computing  $srs^{-1}$  only; that is fine, but the index-2 argument is faster.

**Problem 7.** (20 points) **The class equation.**

- (a) For a finite group  $G$ , choosing representatives  $g_1, \dots, g_m$  of the conjugacy classes of size greater than 1,

$$|G| = |Z(G)| + \sum_{i=1}^m |G : C_G(g_i)|,$$

where  $Z(G)$  collects exactly the elements whose class has size 1, and  $|G : C_G(g_i)|$  is the size of the class of  $g_i$  (orbit–stabilizer for the conjugation action).

- (b)  $Z(D_8) = \{1, r^2\}$ , and the remaining classes are  $\{r, r^3\}$ ,  $\{s, sr^2\}$  and  $\{sr, sr^3\}$ . So

$$8 = 1 + 1 + 2 + 2 + 2.$$

- (c)  $Z(A_4) = 1$ . The three double transpositions form a single class of size 3, and the eight 3-cycles split into two classes of size 4 (the  $S_4$ -class of 3-cycles has 8 elements and  $|C_{A_4}((1\ 2\ 3))| = 3$ , so each  $A_4$ -class has  $12/3 = 4$  elements). So

$$12 = 1 + 3 + 4 + 4.$$

- (d)  $Z(S_4) = 1$ , and the classes are the cycle types: identity (1), transpositions (6), double transpositions (3), 3-cycles (8), 4-cycles (6). So

$$24 = 1 + 6 + 3 + 8 + 6.$$

**Grading note.** Part (c) is the discriminating one: the split of the 3-cycles into two  $A_4$ -classes is the point. A student who writes  $12 = 1 + 3 + 8$  has given a correct partition of 12 into class-*type* sizes but the wrong class equation; award 2 of 5. In (b) and (d) the counts must be class sizes, each dividing  $|G|$  — a quick sanity check students should perform themselves.

**Problem 8.** (20 points) **Isomorphic or not.**

- (a) Both have order 60. Elementary divisors:  $\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$  gives 2, 3 and 2, 5, i.e.  $\{2, 2\}$  at  $p = 2$ ,  $\{3\}$  at  $p = 3$ ,  $\{5\}$  at  $p = 5$ ;  $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/30\mathbb{Z}$  gives 2 and 2, 3, 5, i.e. the same lists. Since the elementary divisors agree, the groups are isomorphic. (Both have invariant factors  $2 \mid 30$ .)
- (b)  $D_{24}$  contains  $r$  of order 12. Every element of  $S_4$  has cycle type  $1^4$ ,  $(2, 1, 1)$ ,  $(2, 2)$ ,  $(3, 1)$  or  $(4)$ , hence order at most 4. An isomorphism preserves orders of elements, so  $D_{24} \not\cong S_4$ .
- (c) An internal semidirect product  $G = H \rtimes K$  requires in particular  $H \cap K = 1$ . But every nontrivial subgroup of  $Q_8$  contains  $-1$ : the subgroups of  $Q_8$  are  $1$ ,  $\{\pm 1\}$ ,  $\langle i \rangle$ ,  $\langle j \rangle$ ,  $\langle k \rangle$  and  $Q_8$ , and every one of them except the trivial subgroup contains  $-1$  (indeed  $-1$  is the unique element of order 2, so any subgroup of even order contains it, and every nontrivial subgroup has even order by Lagrange). Hence for proper nontrivial  $H, K \leq Q_8$  we always have  $-1 \in H \cap K \neq 1$ , and  $Q_8$  is not such a semidirect product.
- (d) Take  $G = \mathbb{Z}/4\mathbb{Z}$  and  $N = \langle \bar{2} \rangle \cong \mathbb{Z}/2\mathbb{Z}$ . Then  $N \trianglelefteq G$  (as  $G$  is abelian) and  $G/N \cong \mathbb{Z}/2\mathbb{Z}$ , but  $N \times (G/N) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$  has no element of order 4, whereas  $G = \mathbb{Z}/4\mathbb{Z}$  does. So  $G \not\cong N \times (G/N)$ .

**Grading note.** Parts (b) and (d) are one-liners using the element-order invariant and should be quick. Part (c) is the interesting one: the key observation is that  $-1$  lies in every nontrivial subgroup, which is a consequence of  $-1$  being the unique element of order 2; award 4 of 6 for that observation alone. Students who instead try to enumerate all homomorphisms  $K \rightarrow \text{Aut}(H)$  will not finish and should be given credit only for correct partial work.